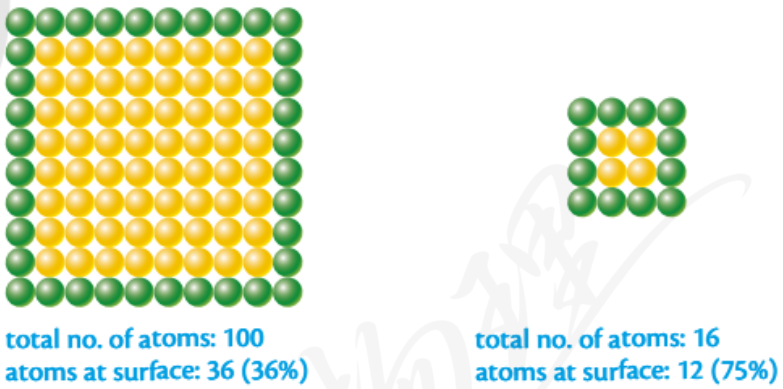
1. Nanoscale materials

A nanometre is one billionth of a metre. Nanomaterials usually have at least one dimension around 1–100 nm. Increased surface-atom fractions and structural or quantum effects can change their electrical, mechanical and optical properties relative to bulk material. Do not attribute every change to surface area alone. Explain the material-specific mechanism.

$$1\,\text{nm} = 10^{-9}\,\text{m}$$

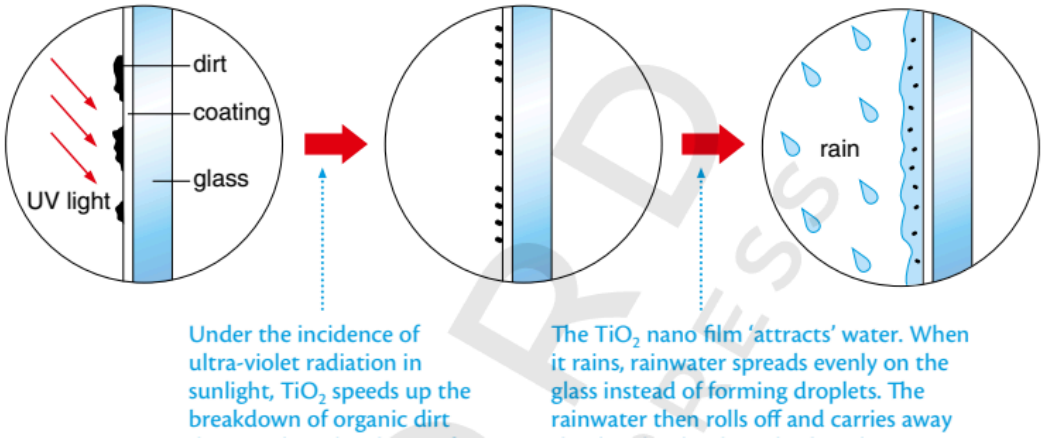
Nanomaterials typically have at least one dimension between 1 and 100 nm.



Smaller particles have a greater fraction of atoms at their surfaces.

2. Self-cleaning coatings and carbon structures

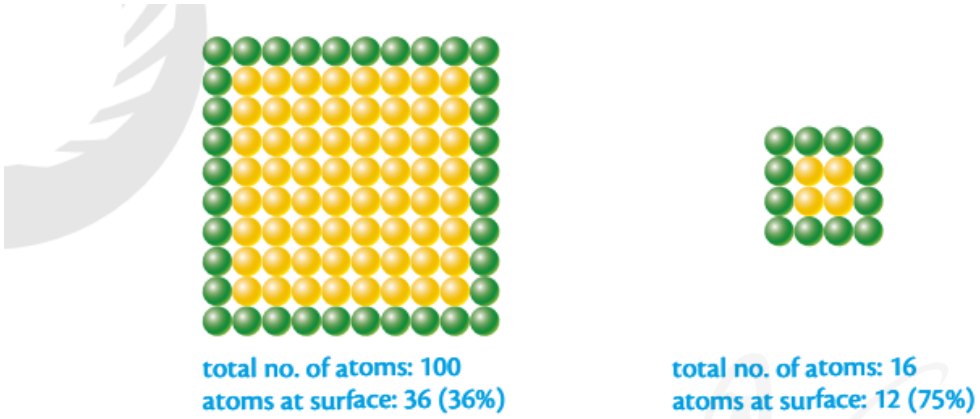
Titanium-dioxide coatings have two cleaning mechanisms. Ultraviolet light assists the breakdown of organic dirt, while a hydrophilic surface lets rain spread and wash it away. The lotus effect instead uses water-repelling droplets. Carbon materials show how structure affects properties: diamond has a three-dimensional covalent network, graphite has sliding layers, graphene is one honeycomb sheet, and nanotubes resemble rolled sheets. They contain the same element but have different atomic arrangements, giving different mechanical and electrical properties.



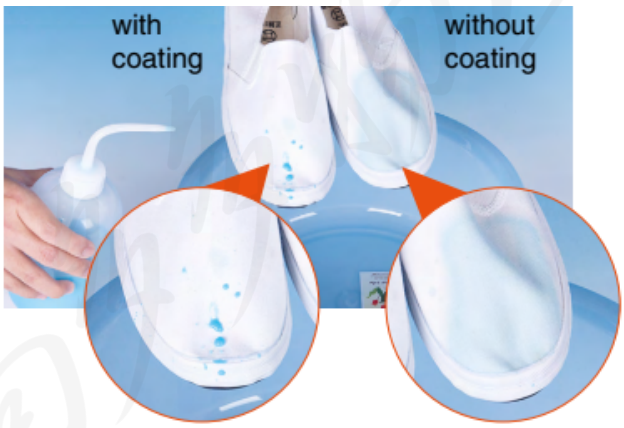
Hydrophilic self-cleaning by a titanium-dioxide coating.

3. Surface effects and the lotus effect

Reducing size increases surface-area-to-volume ratio and the fraction of surface atoms available to interact with the surroundings. A lotus leaf's nanoscale surface structures repel water, allowing rolling droplets to carry dirt away.



Smaller particles have a greater fraction of atoms at their surfaces.



The water-repelling lotus surface lets droplets roll away carrying dirt.

4. Rayleigh's criterion and resolving power

A finite aperture forms a diffraction pattern for each point. Rayleigh's limit occurs when one central maximum coincides with the other's first minimum. For a circular aperture, shorter wavelength or greater diameter reduces the minimum resolvable angle, improving resolution. Resolving two nearby points differs from merely magnifying their image.

$$\theta_R \approx 1.22\frac{\lambda}{d}$$

Rayleigh criterion for a circular aperture. θ_R: minimum resolvable angular separation (rad); λ: wavelength (m); d: aperture diameter (m), or pupil diameter for the eye.

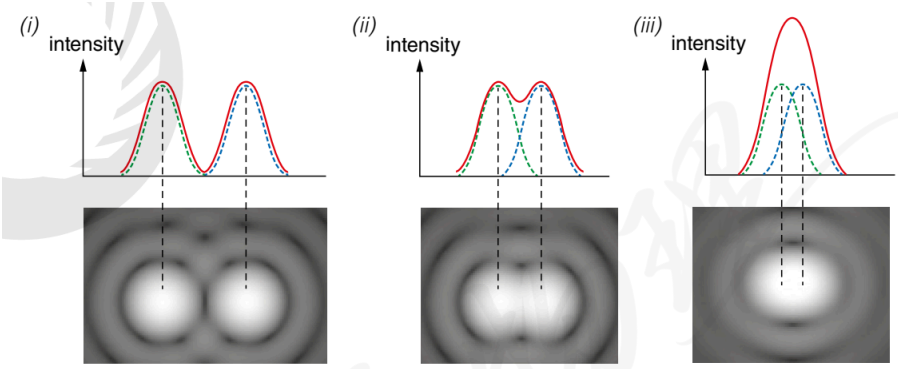


Fig 3.2d *According to Rayleigh criterion, two objects are (i) resolved clearly, (ii) just resolved, and (iii) not resolved.*

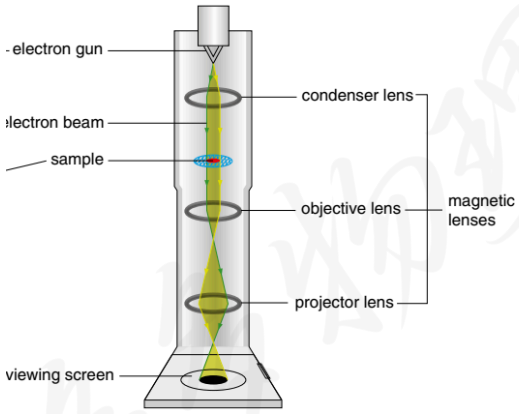
Compare clearly resolved, just resolved and unresolved pairs. At the Rayleigh limit, one central maximum falls on the first minimum of the other pattern.

5. Components of a transmission electron microscope

A transmission electron microscope first uses an electron gun to produce a beam. Condenser magnetic lenses focus it onto a thin sample. Transmitted electrons pass through objective and projector lenses to form an enlarged image, recorded on a screen or camera. Different parts of the sample transmit electrons differently, creating contrast. Magnetic lenses change electron directions using magnetic fields.

6. Transmission electron microscopy

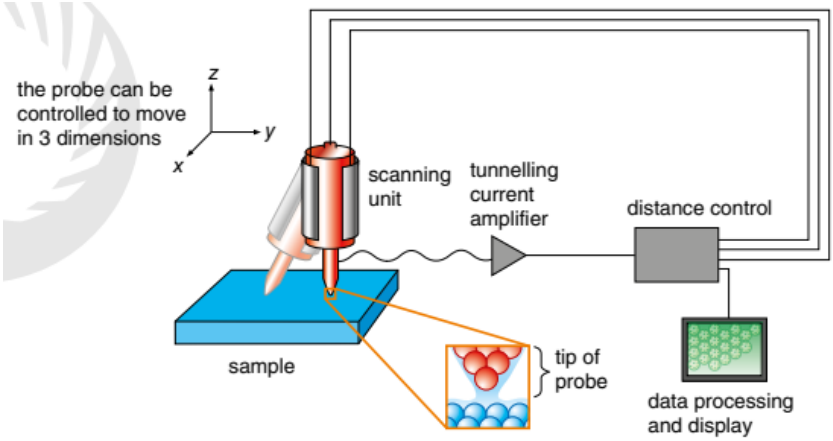
A transmission electron microscope passes fast electrons through a thin sample. Transmission differences create contrast, and magnetic lenses focus and magnify the beam. Higher accelerating voltage increases momentum and reduces de Broglie wavelength, improving the wavelength-related resolution limit. Actual resolution is also limited by lens and specimen effects.



Follow the electron beam through the gun, magnetic lenses and sample to the imaging screen.

7. Scanning tunnelling microscopy

A conducting tip near a conducting sample detects a tunnelling current highly sensitive to separation. Scanning at fixed height while recording current, or adjusting height to keep current constant, reveals surface detail. The image is reconstructed from scanning and tunnelling signals.



A scanning tip detects tunnelling current across a very small gap to obtain surface information.